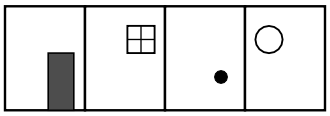
Preview: y2003_q12 (v1)

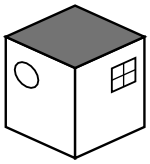
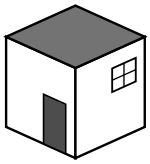
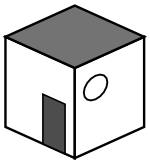
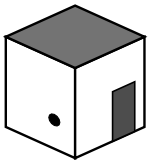
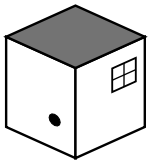
Folding a net in your head: neighbours stay neighbours, in the same order, marks keep their side

Part A: Each question is worth 3 points

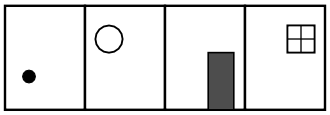
1. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.



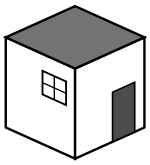
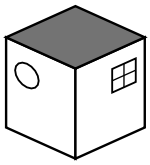
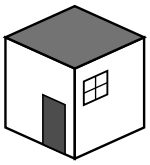
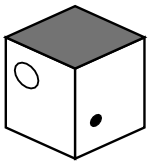
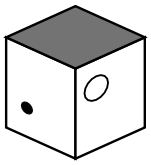
Which house is it?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

2. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.

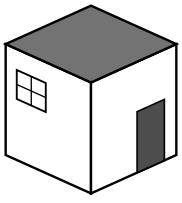


Which house is it?

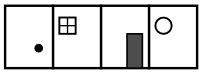
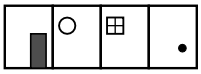
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

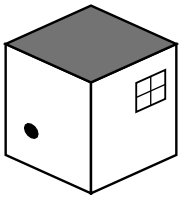
3. A strip of four walls is cut out and folded into this house, with the pictures on the outside. A flat roof is put on top.



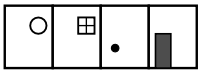
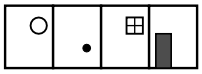
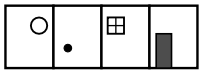
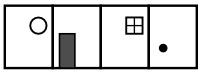
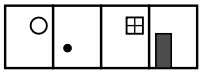
Which strip was it?

A. ***B.***  ***C.*** ***D.***  ***E.***

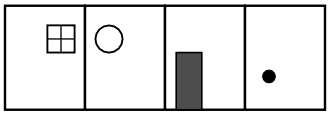
4. A strip of four walls is cut out and folded into this house, with the pictures on the outside. A flat roof is put on top.

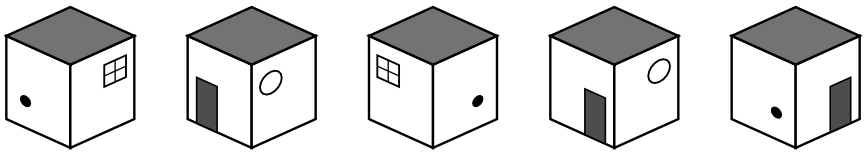


Which strip was it?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

5. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.



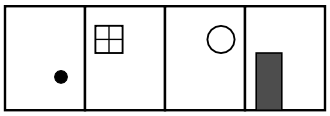


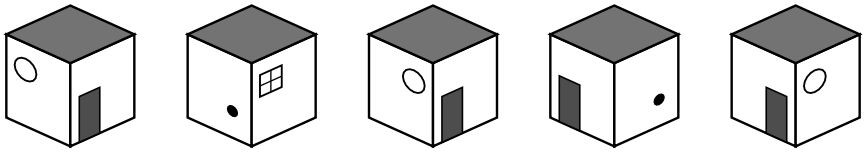
A B C D E

How many of the houses A–E can be made from the strip?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

6. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.





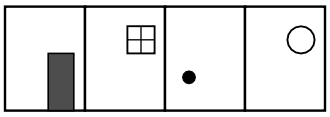
A B C D E

How many of the houses A–E can be made from the strip?

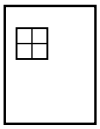
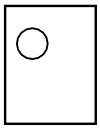
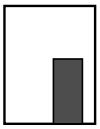
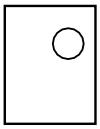
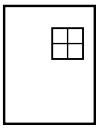
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

Part C: Each question is worth 5 points

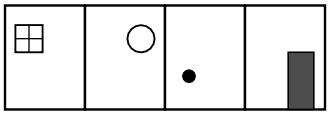
7. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top. You stand outside the house at a corner. The wall with the dot (a doorbell) is on your left.



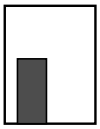
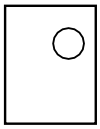
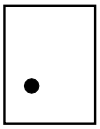
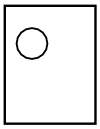
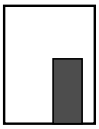
Which wall do you see on your right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

8. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top. You stand outside the house at a corner. The wall with the square window is on your left.



Which wall do you see on your right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Answer Key

1. **B**	2. **E**	3. **B**	4. **E**	5. **B**	6. **D**
7. **D**	8. **B**				

Solutions

1. B — Walls next to each other on the strip meet at a corner, in the same order: the left one stays on the left. Only house B shows two neighbouring walls in the right order with every mark on the correct side. [2003 Q12 · which_house]

2. E — Walls next to each other on the strip meet at a corner, in the same order: the left one stays on the left. Only house E shows two neighbouring walls in the right order with every mark on the correct side. [2003 Q12 · which_house]

3. B — The left wall of the house must come right before the right wall on the strip (or the right wall is first and the left wall last), with each mark on the same side. Only strip B works. [2003 Q12 · which_strip]

4. E — The left wall of the house must come right before the right wall on the strip (or the right wall is first and the left wall last), with each mark on the same side. Only strip E works. [2003 Q12 · which_strip]

5. B (1) — A house is right if its two walls are neighbours on the strip, in the same order, with the marks on the correct side. Right: A, so 1. [2003 Q12 · how_many_houses]

6. D (3) — A house is right if its two walls are neighbours on the strip, in the same order, with the marks on the correct side. Right: B, C, D, so 3. [2003 Q12 · how_many_houses]

7. D — Folded with the pictures outside, the strip keeps its order: on your right is the next wall on the strip. Its mark stays on the same side, so it is picture D. [2003 Q12 · right_neighbour]

8. B — Folded with the pictures outside, the strip keeps its order: on your right is the next wall on the strip. Its mark stays on the same side, so it is picture B. [2003 Q12 · right_neighbour]